

Chapter 41

L. 1.

$\Rightarrow$ monomial maps with codomain $\mathbb{P}^{n_1-1} \times \dots \times \mathbb{P}^{n_e-1}$

$$A_{\bullet} = (A_1, \dots, A_e), \quad A_i \in \mathbb{Z}^{d \times n_i}$$

$$\bar{\Phi}_{A_{\bullet}}: (\mathbb{C}^*)^d \longrightarrow \mathbb{P}^{n_1-1} \times \dots \times \mathbb{P}^{n_e-1}$$

$$t = (t_1, \dots, t_d) \longmapsto (\bar{\Phi}_{A_1}(t), \dots, \bar{\Phi}_{A_e}(t))$$

(all $\overline{\text{im}(\bar{\Phi}_{A_{\bullet}})} = X_{A_{\bullet}}$ a multiproj. variety.

$X_{A_{\bullet}}$ is an irreducible variety and defined by homogeneous prime ideal $I(X_{A_{\bullet}})$ in

$\mathbb{Z}^e$ -graded polynomial ring $\mathbb{C}[\underbrace{x_{11} \dots x_{1n_1}}, \dots, \underbrace{x_{e1} \dots x_{en_e}}]$
 $\deg = (1, 0, \dots, 0) \quad \deg = (0, \dots, 0, 1)$

A homogeneous polynomial $g(x)$ in this ring is

multihomogeneous: $\mathbb{Z}^e$ -grading on $\mathbb{C}[x_1, x_2, \gamma_1, \gamma_2]$

$$g(x) = x_1 \gamma_1 + x_2 \gamma_1 = \deg(1, 1) + \deg(1, 1)$$

$$h(x) = x_1^2 \gamma_2 + x_2 \gamma_2 = \deg(2, 1) + \deg(1, 1)$$

$\nwarrow$ not multihomogeneous.

Cayley configuration: $(d+e) \times (u_1 + \dots + u_e)$ matrix

$$\text{Cay}(A_\bullet) = \begin{pmatrix} A_1 & A_2 & \dots & A_e \\ 1 \dots 1 & 0 \dots 0 & & 0 \dots 0 \\ & 1 \dots 1 & & \vdots \\ & & & 1 \dots 1 \end{pmatrix}$$

Theorem: The multihom. prime ideal $I(X_{A_\bullet})$ of $X_{A_\bullet}$ is the toric ideal $I_{\text{Cay}(A_\bullet)}$.

Proof: $Y_{\text{Cay}(A_\bullet)} = \hat{X}_{A_\bullet} \subset \mathbb{C}^{u_1} \times \dots \times \mathbb{C}^{u_e}$.
 $\hat{X}$ is the affine (multi) cone of X
 $\Rightarrow$ one scaling variable per block

Here $\hat{X}_{A_\bullet}$ is the affine variety obtained by taking the closure of all points

$$((x_{m1}, \dots, x_{m n_1}), \dots, (x_{e1}, \dots, x_{e n_e})) \in \overline{\mathbb{C}^k \setminus \{0\}}$$

s.t.

$$((x_{m1}, \dots, x_{m n_1}), \dots, (x_{e1}, \dots, x_{e n_e})) \in X_{A_\bullet}$$

Then $I(X_{A_\bullet}) = I(\hat{X}_{A_\bullet})$

Example 4.1.4 + Exercise 4.1.3

Corollary: $\dim(X_{A_\bullet}) = \text{rank}(\text{Cay}(A_\bullet)) - e$

$\rightarrow$ analogous to 1.3.18

(1)

Proposition: $T \in \mathbb{Q}^{d \times d}$, $m \in \mathbb{Q}^d$, $\det(T) \neq 0$

and s.t. $T \cdot A_i + m_i \in \mathbb{Q}^{d \times n_i}$

$\Rightarrow$ affine transforms do not change $X_{A_\bullet}$

Then $T \cdot A_\bullet + m_\bullet$ satisfies $X_{A_\bullet} = X_{T \cdot A_\bullet + m_\bullet}$

Proof (Ex. 4.1.7):

For an affine variety X_A we know that the variety is invariant under transforms that leave the row space of A invariant. The reason is that the ideal is defined via the $\ker(A)$ which would also be invariant. Then we know that $I(\hat{X}_{A_\bullet}) = I(X_{A_\bullet})$. So we only need to show that $\text{row}(\text{Cay}(A_\bullet)) = \text{row}(\text{Cay}(T \cdot A_\bullet + m_\bullet))$.

We can consider the blocks separately

Two matrices A, B have the same row span if $M \cdot A = B$.

Here choose $M = \begin{pmatrix} T & \begin{pmatrix} m_1 & \dots & m_1 \\ \vdots & & \vdots \\ m_d & & m_d \end{pmatrix} \\ 0 & \mathbb{I} \end{pmatrix}$

s.t.

$$M \cdot \text{Cay}(A_\bullet) = \text{Cay}(T \cdot A_\bullet + m_\bullet)$$

Corollary $\dim X_{A_\bullet} = \text{rank}(\mathbb{Z}' A_\bullet)$ (2)

Proof (Ex. 4.1.9.): $A \in \mathbb{Z}^{d \times n}$

$$\mathbb{Z}' A = \left\{ \sum_{i=1}^n c_i a_i : c_i \in \mathbb{Z}, \sum c_i = 0 \right\} \subset \mathbb{Z}^d$$

Proposition: (SNF)

$\Rightarrow$ we can find 1-to-1 parametrization $\bar{A}_\bullet$

s.t. $X_{A_\bullet} = X_{\bar{A}_\bullet}$

4.2

Segre embedding:

$$\sigma: \mathbb{P}^{n_1-1} \times \dots \times \mathbb{P}^{n_e-1} \longrightarrow \mathbb{P}^{n_1+\dots+n_e-1}$$

$$((x_{11}, \dots, x_{1n_1}), \dots, (x_{e1}, \dots, x_{en_e})) \mapsto (x_{1i_1} x_{2i_2} \dots x_{ei_e})_{(i_1, \dots, i_e)}$$

Construct $A_{\sigma} \in \mathbb{Z}^{dx_{n_1+\dots+n_e}}$. Label the $n_1+\dots+n_e$ columns by $(i_1, \dots, i_e)$ with $1 \leq i_j \leq n_j$

The ordering of columns matches the ordering of coordinates σ .

The column $(i_1, \dots, i_e)$ is $a_i^{(1)} + \dots + a_{ie}^{(e)}$ with $a_{ij}^{(j)}$ is the i_j -th column of A_j

Then we have an isomorphism

$$X_{A_0} \subseteq X_{A_{\sigma}} \subset \mathbb{P}^{n_1+\dots+n_e-1} \text{ via } \bar{\Phi}_{A_{\sigma}} = \sigma \circ \bar{\Phi}_{A_0}$$

Example 4.2.2.

Definition: Minkowski sum of sets $S_1, S_2 \subset \mathbb{R}^d$

is

$$S_1 + S_2 = \{s_1 + s_2 : s_1 \in S_1, s_2 \in S_2\}$$

Define $A_i \subset \mathbb{Z}^d$ the set of columns of A_i and define the Minkowski sum $A_+ = A_1 + \dots + A_e$

$\Rightarrow A_+$ equals the set of columns of A_+

Corollary: $A \in \mathbb{Z}^{d \times |A_+|}$ the matrix whose columns are the points in A_+ .

Then

$$X_{A_+} \subseteq X_{A_+} \subseteq X_{A_+} \subseteq \mathbb{P}^{|A_+|-1}$$

Proposition:

(3)

$$\begin{aligned} \dim(X_{A_+}) &= \dim(\text{Conv}(A_+)) \\ &= \dim(\underbrace{\text{Conv}(A_1) + \dots + \text{Conv}(A_e)}_{\text{convex polytope}}) \end{aligned}$$

convex polytope

Example 4.2.8.